Benchmarking LLMs via Uncertainty Quantification

Fanghua Ye Mingming Yang Jianhui Pang Longyue Wang
Derek F. Wong Emine Yilmaz Shuming Shi Zhaopeng Tu

Tencent AI Lab · UCL · University of Macau

NeurIPS 2024 – Datasets and Benchmarks Track

Objective

Problem

Current LLM leaderboards (HuggingFace, OpenCompass . . .) rank models by **accuracy only**.

Two models with *equal accuracy* may differ greatly in **uncertainty** – yet benchmarks treat them identically.

Goal

Integrate **uncertainty quantification** into LLM benchmarking via **conformal prediction** – a statistically rigorous, distribution-free, model-agnostic method.

Objective

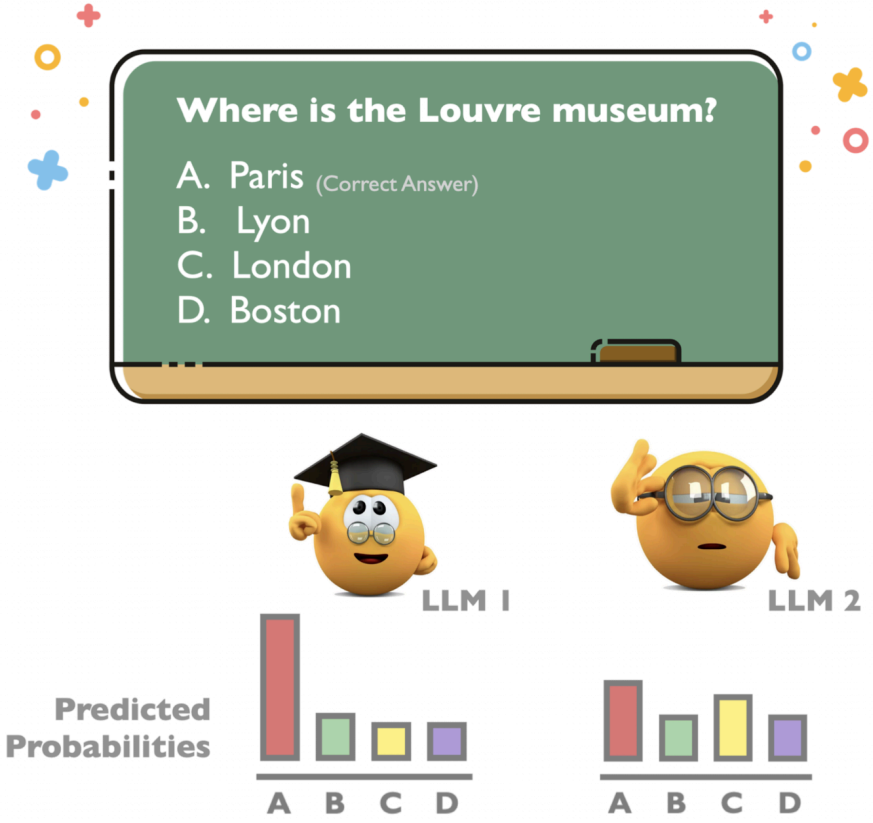

Figure: An illustration of two LLMs accurately predicting the true answer (with option A possessing the highest probability, but showing different levels of uncertainty. Note that when both LLMs predict a wrong answer, they may also display different levels of uncertainty.

Methodology – Conformal Prediction

Coverage Guarantee

For test instance X_t with true label Y_t, CP outputs a **prediction set** $\mathcal{C}(X_t) \subseteq \mathcal{Y}$ such that:

$$P\big(Y_t \in \mathcal{C}(X_t)\big) \;\geq\; 1 - \alpha$$

α = user-specified error rate (e.g. $\alpha = 0.1 \;\Rightarrow$ **90 %** coverage guarantee).

Uncertainty Measure

Set Size (SS) = average $|\mathcal{C}(X_t)|$ over all test instances.

Small SS $\Rightarrow$ **confident model** Large SS $\Rightarrow$ **uncertain model**

Methodology – Score Functions

LAC – Least Ambiguous Classifier

$$s(X, Y) = 1 - f(X)_Y$$

Uses softmax score of the true label only.

$\Rightarrow$ tends to smaller sets.

APS – Adaptive Prediction Sets

$$s(X, Y) = \sum_{\substack{Y' \in \mathcal{Y}: \\ f(X)_{Y'} \geq f(X)_Y}} f(X)_{Y'}$$

Accumulates ranked softmax scores.

$\Rightarrow$ better on hard instances.

Final metrics averaged over **LAC** *and* **APS**.

Algorithm – CP for LLMs

Algorithm 1: Conformal Prediction for LLMs Input: $\mathcal{M}$, $\mathcal{D}$, α Output: Acc, SS, CR

```
for each prompt strategy p ∈ {base, shared, task} do                    // 3 strategies
  for each ICL setting k ∈ {0-shot, few-shot, CoT} do                   // 3 ICL modes
    P(X) ← Softmax(logits_M(X_{p,k}))  for all X ∈ D                    // query LLM
    D → D_cal (50%) + D_test (50%)                                      // random split
    for each s ∈ {LAC, APS} do                                          // 2 score fns
      s_i ← s(X_c^(i), Y_c^(i))  for all i ∈ D_cal                      // nonconformity scores
      q̂ ← Quantile({s_i}, ⌈(n+1)(1−α)⌉ / n)                             // adaptive threshold
      C(X_t) ← {Y' ∈ Y : s(X_t, Y') ≤ q̂}  for all X_t ∈ D_test          // prediction sets
      if C(X_t) = ∅ then keep arg max_{Y'} P(X_t)_{Y'}
return avg over p, k, s:  Acc, SS, CR
```

Methodology – Evaluation Metrics

Metric	Formula	Meaning
Acc	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_p = Y_t]$	Prediction accuracy
SS	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \lvert\mathcal{C}(X_t)\rvert$	Uncertainty (↑ more uncertain)
CR	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_t \in \mathcal{C}(X_t)]$	Coverage ($\geq 1-\alpha$ guaranteed)

Results – Coverage Rate CR (%) per Task

Source: Table 1 – mean of LAC & APS Target: CR $\geq$ 90%

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	92.58	95.20	95.29	91.92	89.71	92.94
Yi-6B	91.30	94.06	92.35	91.36	91.38	92.09
Gemma-7B	93.57	94.16	92.13	91.59	90.70	92.43
Mistral-7B	93.00	92.91	89.94	91.02	92.05	91.78
Llama-2-13B	92.59	93.15	90.50	90.92	90.82	91.60
Qwen-7B	92.79	94.02	91.53	92.43	89.56	92.07
InternLM-7B	90.68	93.28	90.10	90.40	90.34	90.96
Llama-2-7B	91.37	90.69	90.97	89.60	90.04	90.53
DeepSeek-7B	91.18	89.95	90.16	90.89	90.21	90.48
MPT-7B	89.79	90.54	90.12	90.80	89.71	90.19
Falcon-7B	90.04	89.95	89.82	90.46	90.71	90.19

Results – Prediction Accuracy Acc (%) per Task

Source: Table 1 – mean of LAC & APS

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	64.25	91.52	91.00	73.90	49.33	74.00
Yi-6B	57.57	85.99	76.50	58.72	66.06	68.97
Gemma-7B	62.24	85.29	73.58	66.79	40.80	65.74
Mistral-7B	60.44	81.94	62.93	53.21	62.16	64.14
Llama-2-13B	52.52	77.23	59.66	52.65	60.05	60.42
Qwen-7B	55.21	83.89	63.70	64.04	32.53	59.87
InternLM-7B	48.37	73.86	46.21	43.72	34.38	49.31
Llama-2-7B	45.60	65.79	43.05	32.61	45.60	46.53
DeepSeek-7B	45.65	65.39	42.66	33.50	42.15	45.87
MPT-7B	29.49	31.69	25.50	24.38	24.86	27.18
Falcon-7B	23.75	24.98	24.91	25.86	24.69	24.84

Results – Prediction Uncertainty SS per Task

Source: Table 1 – mean of LAC & APS Lower SS = more certain

LLM	QA	RC	CI	DRS	DS	Avg
Qwen-14B	2.80	1.74	2.02	1.94	2.37	2.17
Yi-6B	3.20	1.92	1.88	2.85	1.96	2.36
Gemma-7B	2.72	1.88	2.04	2.14	3.11	2.38
Mistral-7B	2.80	1.75	2.48	2.71	2.40	2.43
Llama-2-13B	3.06	2.24	2.72	2.55	2.24	2.56
Qwen-7B	3.26	2.15	2.28	2.51	2.92	2.63
InternLM-7B	3.49	2.19	3.28	3.63	4.47	3.41
Llama-2-7B	3.20	2.39	3.27	3.26	3.30	3.09
DeepSeek-7B	3.34	2.77	3.06	3.40	3.08	3.13
MPT-7B	3.53	3.46	3.60	3.59	3.66	3.57
Falcon-7B	3.90	3.60	3.66	3.64	3.92	3.75

Results – Three Key Findings

Finding 1: Higher Accuracy $\neq$ Lower Uncertainty

InternLM-7B outperforms MPT-7B by **+19.34 %** Acc on DRS, yet shows **higher** SS (3.63 vs. 3.59). Rankings by Acc and SS diverge across *all* tasks.

Finding 2: Larger $\neq$ More Certain

Qwen 1.8B→14B: Acc and certainty both improve.
Qwen 14B→72B: Acc rises, but SS **increases** on RC & DRS – non-monotonic.

Finding 3: Instruction Finetuning Raises Uncertainty

Chat-V1 and Chat-V2 show **consistently higher SS** than base models across Llama-2, Yi, DeepSeek, and Falcon series.

Results – CP vs. Perplexity (InternLM-7B)

Table 2: Coverage Rate (%) & Set Size – target CR $\geq$ 90 %

	CR (%)		SS	
Task	**CP**	**PPL**	**CP**	**PPL**
QA	**90.68**	83.44	3.49	2.89
RC	**93.28**	95.48	2.19	2.39
CI	**90.10**	96.25	3.28	3.97
DRS	**90.40**	86.80	3.63	3.42
DS	**90.34**	87.13	4.47	4.33
Avg	**90.96**	89.82	3.41	3.40

PPL **fails** 90 % on QA, DRS, DS. CP **always meets** the coverage guarantee.

Results – ECE Comparison (InternLM-7B)

Table 3: Expected Calibration Error (%) – lower is better

Task	CP	Entropy	$P_{\max}$
QA	15.83	15.83	15.83
RC	**1.33**	1.32	1.41
CI	**3.16**	3.45	3.75
DRS	**12.11**	12.40	12.45
DS	**9.30**	9.30	9.62
Avg	**8.35**	8.46	8.61

CP achieves the **lowest average ECE** across all methods.

Our Experiment – Qwen3.5-4B on MMLU

Setup

- Model: `qwen3.5:4b` via Ollama (local inference)
- Dataset: MMLU ($n = 200$ test samples, 1-shot, base prompt)
- Logits via continuation logprob scoring (6 API calls/question)
- CP parameters: $\alpha = 0.1$, cal/test split 50 %

Results

Model	**Acc (%)** $\uparrow$	**SS** $\downarrow$	**CR (%)** ≥ 90
Qwen3.5-4B	53.40	3.96	**93.69**

CR $\geq$ 90 % target **met**. SS $= 3.96$ reflects moderate uncertainty on MMLU.

Conclusion & Future Work

Conclusion

- Accuracy alone is **insufficient** for LLM evaluation.
- Conformal prediction offers a **rigorous, model-agnostic** uncertainty measure.
- New **UAcc** metric jointly captures accuracy and certainty.
- Three findings: Acc $\neq$ certainty; larger $\neq$ more certain; finetuning raises uncertainty.

Future Work

- Extend CP to **free-form text generation** beyond MCQA.
- Apply to **multi-modal** and **closed-source** LLMs (GPT-4 etc.).
- Integrate uncertainty into **RAG** and **in-context learning** pipelines.